

Graph-Guided PMX for Constrained Multi-Objective Operating Theatre Scheduling: A Factorial Evaluation within NSGA-II

Masrie Bin Bukhori

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Abstract

Integrated operating theatre scheduling is a constrained multi-objective combinatorial optimisation problem in which surgery sequencing interacts with operating-theatre capacity, surgical-team availability, specialty/equipment eligibility, and downstream ICU and ward demand. This study investigates whether resource-conflict structure can be injected directly into permutation-based evolutionary search. Two problem-tailored components were developed within an NSGA-II framework: Graph-Guided PMX (Graph-PMX), which biases PMX recombination toward resource-conflict information, and Graph-Guided Local Search (GLS), which preferentially explores swaps involving highly conflicted surgeries. A 2x2 factorial design over Small, Medium, and Large instances isolated their individual and interaction effects under an equal 1,500-evaluation budget with 30 independent runs per configuration. The factorial main effect of Graph-PMX reduced mean IGD+ by 17.56%, 19.40%, and 5.21% on Small, Medium, and Large respectively, while GLS generally worsened convergence under the fixed budget. Graph-PMX was therefore retained as the final method. At official scale, Graph-PMX NSGA-II was evaluated for 30 matched-seed runs against standard NSGA-II, MOEA/D, and Pareto Local Search. Relative to standard NSGA-II, Graph-PMX increased mean hypervolume by 4.02% on Medium and 4.32% on Large, both Holm-corrected significant with large paired effects. On the unseen Hidden instance, hypervolume improved by 2.57% without statistical significance, while IGD+ decreased by 22.70% and spread increased by 21.68%, both significant with large effects. The results indicate that resource-conflict information is most useful when injected during recombination, whereas local refinement can consume evaluations without improving population-level Pareto quality. The study therefore supports selective, evidence-driven use of problem structure rather than assuming that every graph-guided component is beneficial.

Keywords: multi-objective optimisation; operating theatre scheduling; NSGA-II; PMX crossover; graph-guided search; memetic algorithm; factorial design

1. Introduction

The scheduling of operating theatre (OT) should be done in a hospital by coordinating elective surgery requests with limited number of operating rooms, surgical staff, compatibility of specialties/equipments and availability of downstream beds. Here, the decision is in essence combinatorial, and a surgery can't be considered on its own without taking into account other surgeries that may be competing for the same theatre, team, equipment, and/or capacity in the ICU and/or the capacity in the ward over the relevant time period. In parallel, a hospital is not going to optimize a single operational goal. There are multiple conflicting objectives in the schedule, such as throughput, resource utilisation and waiting-time equity, which makes a Pareto formulation more appropriate than a single weighted objective (Cardoen, Demeulemeester and Beliën, 2010; Fügener et al., 2014; Guido and Conforti, 2017; Xue, Li and Zhang, 2025).

Multi-objective evolutionary algorithms, which are able to store multiple trade-off solutions in one run, are appealing in this context because of its population-based approach. But, the success of an evolutionary algorithm also requires a way of representing and recombining candidate schedules. The supplied structure was a list of surgery indices, and the candidate was greedily converted to a real OT-day-team schedule. Standard Partially Mapped Crossover (PMX) maintains permutation validity without explicitly incorporating external information relating to those surgeries that are strongly competing for hospital resources. Where one pair of surgeries have little or no interaction, and another pair have the same theatre, team, equipment and beds, both pairs of surgeries are treated through PMX's segment and mapping mechanism. Structure-aware variation (Goldberg and Lingle, 1985; Larrañaga et al., 1999; Tinós, Whitley and Chicano, 2015; Chitty, Yates and Keedwell, 2022) is motivated by the need to distinguish representation-valid and problem-informed inheritance.

The motivation for the central research question of this study (Can explicit resource-conflict information be used to enhance multi-objective evolutionary search for constrained OT scheduling?) comes from this mismatch between the representations. The study does not propose to replace NSGA-II, but rather is an extension of its variation and exploitation mechanisms by adding a Resource Conflict Graph. Graph-PMX employs conflict structure to control the locations of PMX recombination, whereas Graph-Guided Local Search (GLS) concentrates on neighbourhood search in

conflict-rich surgeries. Each element of the design is designed to be "modular": i.e., it can be turned on or off without altering the goals, the decoder, the mutation operator, or the selection mechanism.

One important methodological component is that both parts are not deemed as beneficial only because they are problem oriented. They are isolated via a 2x2 factorial design matching seeds and holding the evaluation budget constant. This enables the study to differentiate between the components which enhance an individual solution and those which enhance the quality of the population-level Pareto set approximation. The factorial evidence demonstrates that Graph-PMX consistently improves, while GLS might be able to locally improve selected parents, but it is less consistent in improving overall convergence when the evaluations of GLS are chosen from the same fixed budget.

After the factorial study, the evidence-based final method, Graph-PMX NSGA-II with standard swap mutation and without GLS is compared with the three required baselines: standard NSGA-II, MOEA/D and Pareto Local Search (PLS). For Small, Medium, Large and unseen Hidden OT instances, a total of thirty matched seed runs are conducted. To evaluate performance, hypervolume (HV), IGD+, spread, runtime, Friedman omnibus tests, Holm-corrected paired Wilcoxon tests and paired rank-biserial effect sizes are used. Final non-dominated solutions also are decoded and audited to ensure the permutation validity and schedule feasibility.

This study has three empirical contributions. First, it constructs two components of search tailored to the problems, each explicitly modeling the surgery-resource conflicts. Secondly, it offers a factorized analysis in a controlled environment and demonstrates that the timing of the graph information injection is critical: the tested local-search allocation is less effective than a crossover. Third, it compares the selected Graph-PMX method in this study with standard baselines and an untuned Hidden instance and demonstrates statistically significant hypervolume improvements on Medium and Large as well as statistically significant IGD+/spread improvements on Hidden. The rest of the paper describes related work, formalizes the optimization problem, introduces the graph-guided method, outlines the experimental protocol, reports results and discusses implications and limitations.

2. Related Work

The permutation encoding is a direct encoding in which each item is represented exactly once and is useful for sequencing decisions. It needs special variation as the usual way of cross-over may create duplicated elements or miss some elements causing invalid permutation. Goldberg and Lingle (1985) proposed PMX to solve this problem, by segment exchange and allele mappings, and later PMX and order-based crossovers and related permutation operators have been confirmed as standard

mechanisms for order-based optimisation in the reviews of Larrañaga et al. (1999). This creates a crucial difference between valid representation and inheritance based on the problem. Canonical PMX relies on relationships induced from selected parental segments and their mappings (as originally formulated it does not consider external information on whether two surgeries conflict on the team, theatre or downstream resource). It is not a lack of the operator PMX being a permutation preserving operator, but it does leave room for other recombination mechanisms that incorporate extra domain information.

Evolutionary variation, however, does not have to be generic as previous research has shown. Nagata and Kobayashi (2013) demonstrate how to highlight structurally significant edges carried over from parent tours, and to merge crossover with local improvement instead of only positional crossover. More directly, Tinós, Whitley and Chicano (2015) partition pseudo-Boolean variables in a variable-interaction graph when performing a crossover, showing that an externally defined interaction structure can have an active influence on whether or not a crossover is done. Likewise, Yan et al. (2020) embed the graph-based information directly in the process of generating offspring of the evolutionary algorithm for a vehicle-routing problem, and integrate this with problem-oriented local refinement. The study is an important precedent for imposing external graph structure on evolutionary variation, though the graph semantics and recombination process are different from what is found in OT resource-conflict-guided PMX. Another example of incorporating information about relationships between elements in recombination comes from the work of Chitty, Yates and Keedwell (2022), who explore edge-quality-aware crossover for the vehicle-routing problem (VRP) based on permutations. More generally, problem knowledge can be used to direct intensification instead of crossover: Lu et al. (2021) use properties of a scheduling problem to generate local-search operators, and Tellache and Kerbache (2023) explicitly model scheduling conflicts in a graph whose edges correspond to jobs that cannot be processed concurrently. Such studies are significant prior art, as they exclude a general claim of novelty in evolutionary optimisation using graph-aware and/or structure-aware and/or knowledge-guided approaches. However, the semantics and operator mechanisms of their representations are significantly different from embedding surgery-level resource conflicts explicitly into the semantics of PMX.

It can be concluded from the literature that there are three complementary foundations: (i) multi-objective evolutionary algorithms (MOEAs) have conveyed valid Pareto-search frameworks for scheduling problems and are suitable for solving complex scheduling problems; (ii) permutation-preserving operators, like PMX, provide a valid recombination mechanism for sequential representations; (iii) problem knowledge can be used to guide recombination or local intensification.

What is less evident in the literature reviewed here is the specific integration of OT resource-conflict information into PMX's recombination decisions. The research opportunity is therefore more narrowly framed as investigating whether a surgery resource-conflict graph can inform PMX recombination within a multi-objective OT scheduling framework while retaining PMX's permutation-validity properties.

3. Problem Formulation

3.1 Decision structure and representation

Suppose that S is the set of elective surgeries, O is the set of operating theatres, T is the set of surgical teams, and D is the set of planning days. A realised schedule assigns each surgery s to at most one tuple (o,d,t) , subject to eligibility and capacity constraints. Not all the entire tuple is directly encoded in the evolutionary chromosome. Rather, a chromosome corresponds to a permutation π of all the surgery indices. The order in which a deterministic greedy decoder tries to place surgeries is determined by its position. It is an indirect representation that ensures a fixed-length permutation and allows for complex resource assignment to be handled by the decoder.

For exposition purposes, introduce a binary x_{sodt} that is equal to 1 if an operation s is performed in the o -th theatre on day d with the t -th team, and 0 otherwise. The schedule should be such that for each surgery, $\sum_{\{o,d,t\}} x_{sodt} \leq 1$. The total length of surgeries scheduled for any combination of OT-day and/or team-day should not exceed the length of that corresponding day. Assigned OT and teams required for supporting the surgery specialty, plus equipment required. A surgery cannot be scheduled before the earliest possible day. Team consecutive-day limits and post-operative ICU/ward occupancy must also remain within the corresponding instance limits and capacities.

3.2 Objectives

All objectives are given in the minimisation form by the supplied evaluator. The first objective is the negative priority-weighted throughput. In theory, the penalties are the case schedule penalty for cases that are late beyond the maximum acceptable wait time, and more stringent penalties for unscheduled cases, which are subtracted from the reward for scheduling high-priority cases. So, lower f_1 values are equivalent to higher effective priority weighted throughput. The second objective, f_2 , quantifies resource-utilisation imbalance with instance-weighted terms related to OT idle capacity and downstream ICU/ward pressure. The waiting-time fairness measurement, f_3 , for priority classes is the dispersion of priority-weighted average waiting times, which is the third objective. The lower the

better for f_2 and f_3 . No single schedule is likely to dominate the trade-off surface because throughput improvements are likely to cause an increase in resource pressures and/or waiting inequality.

3.3 Feasibility by construction

The constraint handling is done in the decoder, not by an explicit pymoo constraint vector. The decoder processes the surgeries one by one and for each surgery, it goes through the candidate days, and only makes an assignment if there is an available OT and team, OT and team daily hours are still within the bounds, the team's consecutive-day limit is not exceeded, and the downstream ICU/ward requirements still satisfy the instance overflow policy. The instance-provided eligibility maps represent the combinations of OT/Teams that are permitted. If no placement is found a request is marked as unscheduled, instead of marking the request as infeasible. Therefore feasibility refers to the realised assignments and bad service is penalised (poor service through lateness and unscheduled demand). The problem class thus does not have any explicit optimisation constraint as hard feasibility is ensured by construction.

4. Graph-Guided Evolutionary Method

4.1 Resource Conflict Graph

The designed components share a common structural layer: Resource Conflict Graph (RCG). Each node represents one surgery request. Pairwise conflict strength rises if two surgeries have the same eligible OTs, the same eligible surgical teams, share surgical equipment, both occupy ICU beds, or both have relatively long ward stays. Equipment overlap is given a higher weight in the implemented conflict score and the rest of the conflict categories are given additive weights. The weighted degree is a simple measure of the competition level a surgery faces in the hospital and is computed as such. Important, the graph is not alternative schedule representation - the chromosome is still a valid permutation. The graph acts as side information that biases search decisions toward structurally difficult surgery requests.

4.2 Graph-Guided PMX crossover

Standard PMX performs the same permutation property while swapping a segment from two parents in exchange and employing a mapping process to fix duplicate symbols. Graph-PMX preserves this permutation-preserving logic but changes the recombination region to be problem-aware. Removing the idea that all possible crossover regions are equally informative, the implementation is biased to recombination events in areas of conflict-rich surgery with RCG conflict information. The local region

around the operator is bounded by a maximum exchange window of 6 positions. The goal is not to make each offspring greedily optimal, rather it is to maximize the chance of recombination altering ordering relationships that are important to the structure, but also remain stochastic and have valid permutations. Unit-level smoke tests on both the design and Hidden instances confirmed that Graph-PMX produced changed children while preserving valid permutations containing every surgery index exactly once.

4.3 Graph-Guided Local Search

Local exploitation is based on the same resource-conflict information used by GLS. It starts with a given permutation, keeps a local Pareto archive, picks one of the most important surgeries with respect to the conflict weight of the degree, finds the surgery in the permutation and constructs swap neighbours in a finite positional neighbourhood (which is set as four in the experiments). The three objectives are applied to candidate schedules when they are decoded and evaluated. Non-dominated local improvements are retained in the local Pareto archive. Thus the design concentrates the evaluations on possible ordering changes that may be important instead of on pairing swaps randomly throughout the entire permutation.

Memetic integration: GLS did not get an unlimited number of extra evaluations. In the case of the factorial experiment, the 1,500 evaluations that were available overall for the experiment were split into 50 evaluations per configuration that is able to handle GLS. NSGA-II, then, needed 1,450 evaluations, while GLS needed at most 50. Local search was called every 5 generations on one of the selected rank-zero solutions and was provided with a budget of 10 evaluations per call. In order to avoid disruption of NSGA-II's diversity mechanism, improved solutions were stored in an external archive and not directly replaced in the evolutionary population.

4.4 Evidence-driven final configuration

The first research idea was to incorporate Graph-PMX and GLS into a graph guided memetic algorithm. As this was a study on the selection of components, the factorial study was deliberately not assumed to be a representation of the bigger framework that needed to be maintained. GLS showed that it could improve individual parents, but its mean main effect on IGD+ was unfavourable across all three design scales, while its HV main effect was negative on Medium and Large. In the case of Graph-PMX, the improvements in IGD+ were consistent and the HV effects were positive on Medium and Large. Thus, the final algorithm was reduced to NSGA-II + Graph-PMX + standard swap mutation. The decision is at the heart of the contribution: problem awareness was helpful, but the usefulness of it hinged on its placement in the search process.

5. Experimental Methodology

5.1 Instances and run protocol

The evaluation was based on the following design instances: Small, Medium, and Large OT (released with the methods) and Hidden (released after method development). The Hidden instance had 200 surgeries and was not further tuned. All reported stochastic comparisons were done using 30 independent runs, with deterministic seed generation. In each comparison, the seed lists were identical in both algorithms or factorial configurations to facilitate paired statistical testing.

5.2 Factorial component study

Graph-PMX and GLS were isolated by a 2x2 factorial design. The four configurations were: A, standard PMX without GLS; B, Graph-PMX without GLS; C, standard PMX with GLS; and D, Graph-PMX with GLS. The population size was set to 50 and each configuration was subject to a common 1,500-evaluation budget per run. In the case of GLS-enabled configurations, 1,450 evaluations were assigned to NSGA-II and up to 50 evaluations were kept for the local search. The study thus consisted of 3 instances x 4 configurations x 30 runs = 360 optimisation runs. Main effects were calculated as the average of the enabled configuration minus the average of the disabled configuration and interaction was calculated as a difference of differences. The main interpretation of the factorial experiment is as evidence for component-selection, and the later budget-matched Graph-PMX-versus-NSGA-II study is the primary confirmatory pairwise comparison.

5.3 Official-scale baseline comparison

A second experiment was performed using Graph-PMX NSGA-II to evaluate its competitiveness against the standard NSGA-II, MOEA/D and PLS after the selection of Graph-PMX NSGA-II. With NSGA-II and Graph-PMX NSGA-II, population 100 was run for 100 generations with 10,000 evaluations per run. PLS was explicitly limited to 10,000 evaluations. The supplied MOEA/D builder used 105 Das-Dennis reference directions for three objectives (13 partitions) over 100 generations, resulting in 10,500 evaluations per run. This 5% difference in the structural budget is not hidden but is reported instead; the main comparison to the NSGA-II remains budget matched.

5.4 Indicators and statistical analysis

In each case an empirical reference front was generated by integrating all the solutions from all the compared algorithms and keeping only the non-dominated ones. A hypervolume reference point was defined as the component-wise maximum objective value plus one, in the pooled data. Hypervolume (higher is better) and IGD+ (lower is better) were used to assess the quality of the runs and spread as it

was implemented by the provided project indicator utility (larger values correspond to wider spread in the current analysis). In the reproducibility outputs, mean, median, standard deviation and interquartile range were used.

For each instance and indicator, the four configurations/algorithms were used with Friedman tests. Where the omnibus test was statistically significant, paired Wilcoxon signed-rank tests were used on matched seeds and adjusted with the Holm correction on the six pairwise comparisons. Paired rank-biserial correlation was used to quantify effect magnitude with negligible (<0.10), small (<0.30), medium (<0.50), and large (≥ 0.50) thresholds. This separation allows for the consideration of statistical evidence and its practical magnitude as opposed to just mean differences.

Table 1. Final experimental protocols.

Study	Algorithms/configurations	Runs	Budget / termination	Purpose
Factorial	A baseline; B Graph-PMX; C GLS; D Graph-PMX+GLS	30 per config/instance	Common 1,500-evaluation budget	Component effects and interaction
Official baseline	NSGA-II; MOEA/D; PLS; Graph-PMX NSGA-II	30 per algorithm/instance	100 generations for EAs; 10,000 eval PLS	Final competitiveness
Hidden	Same four final algorithms	30	Same official settings; no retuning	Generalisation check

6. Results

6.1 Factorial component effects

The designed contrast generated by the factorial study was very obvious. The mean IGD+ was shown to be decreased by 17.56%, 19.40% and 5.21% for Small, Medium and Large respectively by the factorial main effect of Graph-PMX. The HV main effect was basically null on Small (-0.21%), but favourable on Medium (+3.88%) and Large (+0.66%). In contrast, GLS increased IGD+ by 2.03%, 3.46%, and 5.41% on Small, Medium, and Large, respectively, and had a near-neutral HV effect on Small and a negative effect on Medium and Large. The interactions tended to be very small in magnitude compared with the Graph-PMX main effect, and in fact do not appear to have led to an additional systematic benefit of Graph-PMX.

This was confirmed by statistical testing. There were no statistically significant differences between the configurations for the Small factorial study. On Medium, the Friedman tests were significant for HV and IGD+; the pairwise Graph-PMX-versus-baseline comparisons remained significant after

Holm correction for HV (approximately $p=0.006$) and IGD+ (approximately $p=0.032$). In the Large group, all three omnibus tests were significant while the direct Graph-PMX versus baseline pairwise differences were not significant after multiplicity correction. The appropriate inference is therefore that Graph-PMX had its strongest and most consistent component-level evidence on Medium, whereas Large showed favourable mean directions without equally strong pairwise statistical evidence.

Table 2. Main and interaction effects from the 2×2 factorial design, expressed as percentages of the baseline mean. Positive changes are favourable for HV and spread, whereas negative changes are favourable for IGD+.

Instance	Metric	Graph-PMX main effect (%)	GLS main effect (%)	Interaction (%)
Small	HV	-0.212	-0.011	-0.082
Small	IGD+	-17.560	+2.025	+4.887
Small	Spread	-0.758	+0.480	-1.038
Medium	HV	+3.882	-0.699	+0.054
Medium	IGD+	-19.402	+3.462	+0.073
Medium	Spread	-1.581	-0.284	-2.558
Large	HV	+0.659	-0.885	-0.131
Large	IGD+	-5.211	+5.407	+0.876
Large	Spread	+2.287	-0.272	-0.351

2x2 Factorial Main Effects (negative IGD+ is favourable)

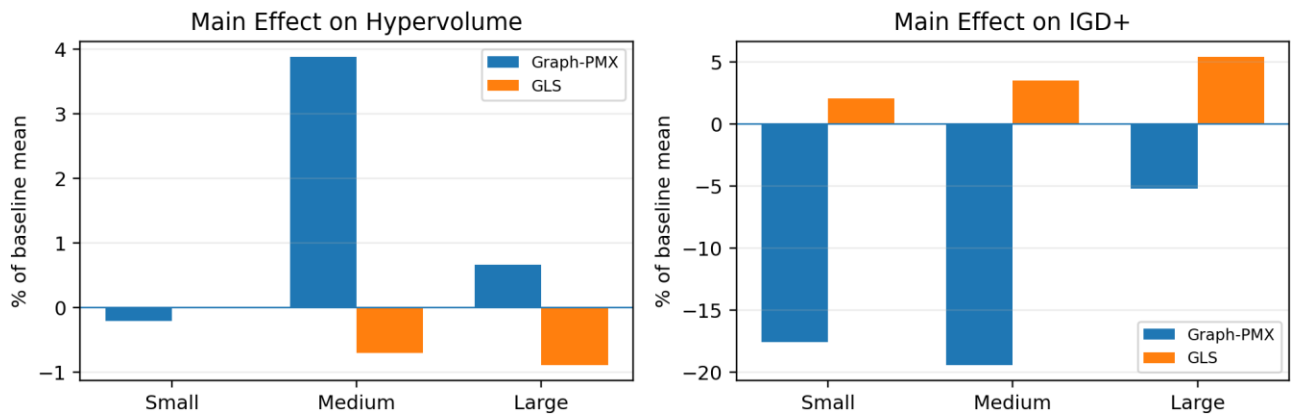


Figure 1. Factorial main effects of Graph-PMX and GLS on hypervolume and IGD+.

6.2 Official-scale comparison with required baselines

The official-scale 30 run results are in Table 3. Graph-PMX NSGA-II performed better than all other algorithms for the mean HV and the spread on all four OT instances. On Small, the HV difference between Graph-PMX and standard NSGA-II was negligible (+0.0045%), while mean IGD+ decreased by 55.45%. On the other hand, the HV differences were materially larger on Medium and Large: +4.02% and +4.32% respectively. After applying Holm correction, both HV differences were statistically significant with large, paired effects. The medium spread increased by quite a bit (+5.95%) and had a medium impact. There was a medium effect size, but not significant, for the IGD+ reduction on Medium (-13.20%) and Large (-10.43%) compared to NSGA-II.

The relationship with MOEA/D is not as straightforward. MOEA/D had very good IGD+ means scores on Medium, Large and Hidden domains, showing its closeness to the parts of the empirical reference front. However, its mean HV and spread were substantially lower than those of both NSGA-II variants, and the fronts returned were generally smaller. It is not to say that the behaviour of Graph-PMX on all indicators was consistently superior, but rather that its behaviour with respect to volume/coverage was superior, whereas MOEA/D had a different convergence-coverage trade-off. PLS was the fastest method on most instances but generally performed worse than Graph-PMX and standard NSGA-II on HV and IGD+ at the larger scales.

Table 3. Performance of algorithms at the official scale (mean +/- sample standard deviation) over 30 independent runs.

Instance	Algorithm	HV	IGD+	Spread	Runtime (s)
Small	NSGA-II	112,054.81 +/- 282.92	0.2012 +/- 0.2837	280.72 +/- 23.56	6.25
Small	MOEA/D	103,342.68 +/- 1,794.90	2.6993 +/- 0.5105	43.53 +/- 7.40	13.78
Small	PLS	109,601.47 +/- 3,495.75	0.9664 +/- 1.0260	262.40 +/- 27.21	4.62
Small	Graph-PMX NSGA-II	112,059.89 +/- 209.24	0.0896 +/- 0.1660	283.04 +/- 17.62	7.64
Medium	NSGA-II	80.41M +/- 3.67M	147.8642 +/- 38.5035	2,575.18 +/- 291.49	34.80
Medium	MOEA/D	57.96M +/- 9.36M	48.2074 +/- 23.9949	164.75 +/- 100.01	47.64
Medium	PLS	60.62M +/- 4.76M	383.8096 +/- 84.7305	1,562.51 +/- 226.41	24.60

Instance	Algorithm	HV	IGD+	Spread	Runtime (s)
Medium	Graph-PMX NSGA-II	83.65M +/- 2.99M	128.3406 +/- 28.3337	2,728.35 +/- 297.86	37.47
Large	NSGA-II	458.34M +/- 25.56M	505.6915 +/- 102.9115	6,511.46 +/- 631.33	87.86
Large	MOEA/D	204.13M +/- 49.44M	121.1145 +/- 63.6138	429.42 +/- 123.06	108.58
Large	PLS	189.23M +/- 34.66M	2,749.8419 +/- 405.8368	2,937.57 +/- 743.48	62.70
Large	Graph-PMX NSGA-II	478.15M +/- 19.18M	452.9278 +/- 120.1367	6,669.78 +/- 709.75	218.52
Hidden	NSGA-II	122.96M +/- 6.23M	177.7363 +/- 50.9779	2,649.82 +/- 445.25	56.45
Hidden	MOEA/D	36.83M +/- 14.86M	78.6366 +/- 61.5625	231.52 +/- 76.18	75.57
Hidden	PLS	56.60M +/- 11.79M	1,308.2548 +/- 286.7243	1,621.65 +/- 492.13	39.58
Hidden	Graph-PMX NSGA-II	126.12M +/- 6.41M	137.3833 +/- 35.8670	3,224.33 +/- 488.64	63.95

6.3 Direct Graph-PMX versus NSGA-II statistical comparison

Budget-matched comparison of Graph-PMX versus NSGA-II compares the impact of using the graph-guided PMX instead of the standard PMX for the official scale. As predicted by the factorial findings, there is no corrected significance for small. Medium and Large show significant HV gains with large effects. A different pattern appears on the Hidden instance: mean HV increased by 2.57% without statistical significance, while IGD+ decreased by 22.70% and spread increased by 21.68%; both latter differences were statistically significant with large effects. Because the Hidden instance was not used for tuning, these results provide evidence that the graph-guided operator transfers beyond the three design instances, although not every indicator improves significantly on every scale.

Table 4. Graph-PMX NSGA-II relative to standard NSGA-II: Holm-corrected paired Wilcoxon signed-rank tests and paired rank-biserial effect magnitudes.

Instance	Metric	Mean change (%)	Holm p	Significant	Effect magnitude
Small	HV	+0.00	0.967674	No	Negligible
Small	IGD+	-55.45	0.140283	No	Medium
Small	Spread	+0.83	0.792159	No	Negligible

Instance	Metric	Mean change (%)	Holm p	Significant	Effect magnitude
Medium	HV	+4.02	0.004735	Yes	Large
Medium	IGD+	-13.20	0.087939	No	Medium
Medium	Spread	+5.95	0.036435	Yes	Medium
Large	HV	+4.32	0.003728	Yes	Large
Large	IGD+	-10.43	0.076721	No	Medium
Large	Spread	+2.43	0.381798	No	Small
Hidden	HV	+2.57	0.104840	No	Medium
Hidden	IGD+	-22.70	0.000872	Yes	Large
Hidden	Spread	+21.68	0.000137	Yes	Large

Graph-PMX NSGA-II: Relative Change from Standard NSGA-II

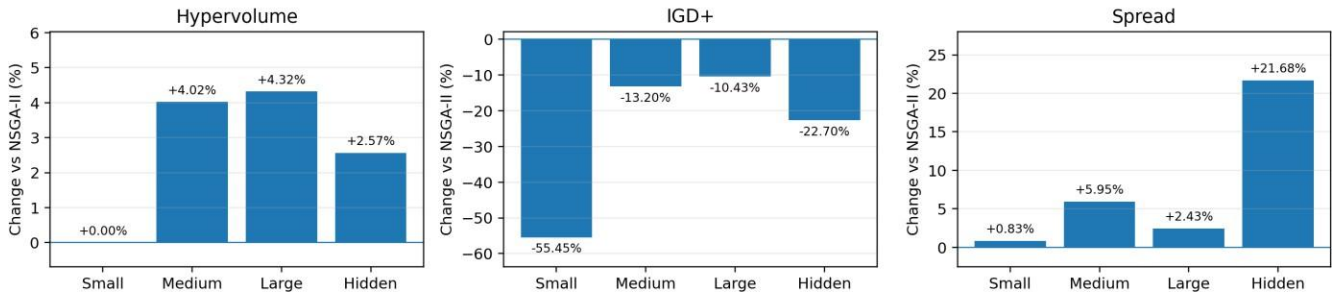


Figure 2. Relative mean performance change of Graph-PMX NSGA-II from standard NSGA-II.

Negative IGD+ changes are favourable.

6.4 Feasibility and final Pareto-set output

In all cases, objective vectors that are the same or duplicate were eliminated from the global non-dominated union and all the run fronts of Graph-PMX were merged. Retained permutations were then decoded into surgery -> OT-day-team schedules and output with objective values and decoder-based feasibility certificates. Every surgery index appeared in each one of the exported permutations, and feasibility-by-construction decoder was used to generate all realised schedules without any duplicate surgery assignments. The final Pareto approximations had 66, 288, 302, and 274 different objective vectors for Small, Medium, Large, and Hidden, respectively. Because the decoder does not attempt a placement that is infeasible, some requests stay unscheduled; this is part of the optimisation trade-off, but not considered a violation of a hard constraint, but penalised by the first objective.

Table 5. Final Graph-PMX non-dominated schedule export and decoder-based feasibility certificate.

Instance	Unique ND solutions	Scheduled surgeries range	Unscheduled range	Decoder-feasible exports
Small	66	18-21	1-4	100%
Medium	288	105-114	6-15	100%
Large	302	228-238	42-52	100%
Hidden	274	152-158	42-48	100%

7. Discussion and Threats to Validity

The most important finding is not that graph information automatically improves an evolutionary algorithm. Rather, the location of graph guidance matters. Graph-PMX modifies information exchange during recombination, where ordering relationships are created and disrupted at population scale. GLS, by contrast, spends a limited evaluation budget exploiting a small neighbourhood around selected solutions. Under the common 1,500-evaluation budget, those local improvements were not consistently strong enough to compensate for evaluations removed from global exploration. This explains why a locally improved parent can coexist with a worse population-level HV or IGD+ outcome.

The scale pattern is also informative. On Small, the search space is sufficiently limited that standard PMX already performs near the same HV level and no corrected pairwise improvement is detected. Medium provides the clearest factorial evidence for Graph-PMX, and official-scale Medium/Large both show significant HV gains over NSGA-II. A plausible interpretation is that conflict-guided recombination becomes more valuable as the number of surgeries and interacting eligibility/resource relations grows. However, this should remain an empirical interpretation rather than a universal claim; only the supplied OT instances were tested.

The multi-indicator results caution against declaring a single universal winner. Graph-PMX maximised HV and spread across the reported OT instances, but MOEA/D achieved the lowest IGD+ on Medium, Large, and Hidden. This suggests that different algorithms emphasise different regions or shapes of the trade-off surface. For decision support, the broader coverage produced by Graph-PMX may be valuable because it preserves a wider set of throughput-resource-fairness compromises, whereas MOEA/D's lower IGD+ may favour proximity to a subset of the empirical reference front. A hospital decision maker would still need to select among Pareto alternatives according to operational priorities.

Several validity threats remain. First, the true Pareto front is unknown, so IGD+ and HV are computed against empirical reference sets derived from the compared runs; improved algorithms can change the reference set itself. Second, the official MOEA/D protocol performs 10,500 evaluations because 105 reference directions are structurally generated, whereas the other final baselines use 10,000. This slightly favours MOEA/D in evaluation count, although the primary Graph-PMX-versus-NSGA-II comparison is exactly matched. Third, feasibility is guaranteed by the supplied greedy decoder, so the proposed crossover does not independently solve constraint handling; its contribution is to ordering search under the same decoder. Fourth, unscheduled surgeries are permitted and penalised rather than treated as infeasible, so solution quality depends partly on the first objective's penalty design.

Fifth, ICU and ward stay lengths are deterministic point estimates. Real postoperative demand is uncertain, and a schedule feasible under deterministic stays may be fragile under stochastic length-of-stay variation. Sixth, the Hidden experiment provides an important unseen-instance check but represents only one additional OT instance. No external scheduling benchmark is claimed in this study, so transferability beyond the OT domain remains untested. Finally, Graph-PMX incurs computational overhead, most visibly on Large where mean runtime rises from 87.86 s for standard NSGA-II to 218.52 s. The operator therefore trades additional computation for improved Pareto volume/coverage, and future work should examine more efficient conflict-score updates or selective invocation.

8. Conclusion and Future Work

This study aimed to investigate the potential of using resource-conflict information to enhance the multi-objective evolutionary search in the integrated operating theatre scheduling. Two graph guided components were designed and tested: graph-guided PMX crossover and graph-guided local search. In the controlled 2x2 factorial study, the contribution of Graph-PMX was more consistent, for example for convergence on Medium level, while GLS mostly used evaluation budget without delivering a similar benefit on the population level. Based on this evidence, the final method was to keep Graph-PMX in NSGA-II and eliminate GLS. When compared across scales, Graph-PMX NSGA-II achieved the best mean hypervolume and spread on Small, Medium, Large and Hidden. Hypervolume gains from the standard NSGA-II were significant and the paired effects were large on Medium and Large. The HV increase was not significant in the unseen Hidden instance, and IGD+ and spread improved significantly with large effects in this instance. In support of a modest conclusion, these results suggest that resource-aware recombination is a valuable extension to this OT scheduling formulation, particularly in the presence of increasing structural complexity, but is not the best baseline on all

measures, and it does not have a negligible runtime overhead. It would be valuable to examine whether the graph guidance can be applied to other scheduling problems and other external benchmarking tasks. Second, the scores of conflicts might not be based on instances but be updated dynamically based on the actual behaviour of the population. Third, local search could be triggered adaptively only when expected marginal benefit exceeds its evaluation cost. Lastly, deterministic ICU/ward demand needs to be extended to scenario-based or stochastic postoperative occupancy, such that not only nominal feasibility, but also robustness is applied for judging Pareto solutions.

Data, Code and Reproducibility

The submission package includes the integrated source code, precise Python requirements, seeded runs per run JSON logs, final (decoded) non-dominated runs schedules, and a standalone analysis python script that generates 480 official runs schedules, empirical reference fronts, descriptive stats, Friedman tests, and Holm corrected posthoc analysis results as outputs from raw logs. A representative analysis can be reproduced with the documented command in the project README, using a marker.

Acknowledgements / AI-use statement

Generative AI tools were used for supporting code debugging, concept clarification, experimental-pipeline checking, language drafting/refinement, and literature discovery. The Related Work section has been created using candidate academic sources identified and mapped by ChatGPT Deep Research. The author did the experiments, read the results, finalized methodological choices, and is responsible for checking, paraphrasing, citing the final version and defending it.

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